

## Answers: Logic Drill #6

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| Section | Topic                    | Questions | Target time  |
|---------|--------------------------|-----------|--------------|
| A       | Rapid Recognition        | 10        | 2 min 30 sec |
| B       | Manipulation Drills      | 10        | 8 min        |
| C       | Substitution & Structure | 8         | 10 min       |
| D       | Challenge                | 5         | 15 min       |

### Section A

1. What are the two essential components required for a proof by mathematical induction?

**Answer:** The Base Case and the Inductive Step

**Method:** A proof by mathematical induction requires: (1) Base Case: establishing  $P(n_0)$  for the initial value  $n_0$ , and (2) Inductive Step: proving  $P(k) \implies P(k+1)$  for all  $k \geq n_0$ .

**Investigate further:** Why does combining the base case and inductive step cover all integers  $n \geq n_0$ ?

2. To prove a statement  $P(n)$  holds for all integers  $n \geq 4$  by induction, what is the base case?

**Answer:**  $P(4)$

**Method:** The base case must be the smallest value of  $n$  in the domain of the statement, which is  $n = 4$ .

**Investigate further:** What happens if you check  $P(1)$  instead when  $P(n)$  is only defined for  $n \geq 4$ ?

3. If  $P(k) \implies P(k+1)$  is proved, but no base case  $P(n_0)$  is established, is  $P(n)$  proved for any  $n$ ?

**Answer:** No

**Method:** Without a base case to anchor the chain of implications, no value of  $P(n)$  is known to be true. An unanchored inductive step proves only conditional relationships, not truth values.

**Investigate further:** Give an example of a false statement  $P(n)$  where  $P(k) \implies P(k+1)$  holds for all  $k$ .

4. What is the inductive hypothesis in a proof of  $\sum_{i=1}^n i = \frac{n(n+1)}{2}$ ?

**Answer:** Assume  $\sum_{i=1}^k i = \frac{k(k+1)}{2}$  for some integer  $k \geq 1$ .

**Method:** The inductive hypothesis is the assumption that the target formula holds for a specific fixed integer  $k$ .

**Investigate further:** What is the next expression you must derive using this hypothesis?

5. True or false: “Mathematical induction can be used to prove statements about all real numbers  $x$ .”

**Answer:** False

**Method:** Mathematical induction applies to discrete countable sets ordered by successor functions (like

integers  $\mathbb{Z}^+$ ), not continuous sets like  $\mathbb{R}$ .

Investigate further: What property of the natural numbers makes mathematical induction valid?

6. In a proof that  $2^n > n^2$  for  $n \geq 5$ , what is the smallest integer  $n$  for which the base case must be checked?

**Answer:**  $n = 5$

**Method:**  $2^5 = 32 > 5^2 = 25$  is true. For  $n \in \{2, 3, 4\}$ ,  $2^n > n^2$  is false ( $4 \leq 4, 8 < 9, 16 \leq 16$ ). Thus  $n = 5$  is the first valid base case.

Investigate further: Why does the inductive step  $2^k > k^2 \implies 2^{k+1} > (k+1)^2$  require  $k \geq 5$ ?

7. What is the negation of the inductive step statement: “For all  $k \geq 1$ ,  $P(k) \implies P(k+1)$ ”?

**Answer:** There exists  $k \geq 1$  such that  $P(k)$  is true and  $P(k+1)$  is false.

**Method:** Negating  $\forall k, (P(k) \implies P(k+1))$  yields  $\exists k, \neg(P(k) \implies P(k+1))$ , which simplifies to  $\exists k, (P(k) \wedge \neg P(k+1))$ .

Investigate further: How does this relate to finding a break in an inductive chain?

8. True or false: “If  $P(1)$  is true and  $P(k) \implies P(k+2)$  for all  $k \geq 1$ , then  $P(n)$  is true for all positive integers  $n$ .”

**Answer:** False

**Method:** This step size of 2 only proves  $P(n)$  for all \*odd\* positive integers  $(1, 3, 5, \dots)$ . Even positive integers remain unproved unless  $P(2)$  is also established.

Investigate further: What extra base case would make  $P(n)$  true for all positive integers?

9. For which positive integers  $n$  is  $2^n > n$  true?

**Answer:** All positive integers  $n \geq 1$

**Method:** For  $n = 1$ ,  $2^1 = 2 > 1$ . Assuming  $2^k > k$ , we have  $2^{k+1} = 2 \cdot 2^k > 2k = k + k \geq k + 1$  for all  $k \geq 1$ . Thus it holds for all  $n \geq 1$ .

Investigate further: Is  $2^n > n^2$  true for all  $n \geq 1$ ?

10. What is the base case needed to prove  $n! > 2^n$  by induction?

**Answer:**  $n = 4$

**Method:** For  $n = 1, 2, 3$ :  $1! = 1 < 2$ ,  $2! = 2 < 4$ ,  $3! = 6 < 8$ . For  $n = 4$ :  $4! = 24 > 16 = 2^4$ . Thus  $n = 4$  is the base case.

Investigate further: What is the ratio  $(k+1)!/2^{k+1}$  in terms of  $k!/2^k$ ?

## Section B

1. Express  $\sum_{i=1}^{k+1} i^2$  in terms of  $\sum_{i=1}^k i^2$ .

**Answer:**  $\sum_{i=1}^{k+1} i^2 = \left(\sum_{i=1}^k i^2\right) + (k+1)^2$

**Method:** Splitting off the last term of the summation gives  $\sum_{i=1}^{k+1} i^2 = \left(\sum_{i=1}^k i^2\right) + (k+1)^2$ .

**Investigate further:** Why is splitting off the final term the fundamental first step in inductive sum proofs?

2. Assuming  $\sum_{i=1}^k i = \frac{k(k+1)}{2}$ , add  $(k+1)$  and simplify to show  $\sum_{i=1}^{k+1} i = \frac{(k+1)(k+2)}{2}$ .

**Answer:**  $\frac{k(k+1)}{2} + (k+1) = (k+1) \left[\frac{k}{2} + 1\right] = (k+1) \left[\frac{k+2}{2}\right] = \frac{(k+1)(k+2)}{2}$

**Method:** Factor out  $(k+1)$ :  $\frac{k(k+1)}{2} + (k+1) = (k+1) \left(\frac{k}{2} + 1\right) = (k+1) \left(\frac{k+2}{2}\right) = \frac{(k+1)(k+2)}{2}$ .

**Investigate further:** How does this complete the inductive step?

3. Prove by induction:  $\sum_{i=1}^n (2i-1) = n^2$  for all positive integers  $n$ .

**Answer:** Proof by induction

**Method:** Base case  $n = 1$ :  $\sum_{i=1}^1 (2i-1) = 1 = 1^2$ , true.

Inductive step: Assume  $\sum_{i=1}^k (2i-1) = k^2$ . Then  $\sum_{i=1}^{k+1} (2i-1) = k^2 + (2(k+1)-1) = k^2 + 2k + 1 = (k+1)^2$ . By induction, the identity holds for all  $n \geq 1$ .

**Investigate further:** What is the geometric visualization of this sum using square grids?

4. Assuming  $3^{2k} - 1$  is divisible by 8, express  $3^{2(k+1)} - 1$  in a form that shows it is also divisible by 8.

**Answer:**  $3^{2(k+1)} - 1 = 9(3^{2k} - 1) + 8$

**Method:**  $3^{2(k+1)} - 1 = 9 \cdot 3^{2k} - 1 = 9(3^{2k} - 1) + 8$ . By IH,  $3^{2k} - 1 = 8m$ , so  $9(8m) + 8 = 8(9m + 1)$ , which is divisible by 8.

**Investigate further:** Can you also write  $3^{2(k+1)} - 1 = (3^{2k} - 1) + 8 \cdot 3^{2k}$ ?

5. Prove by induction:  $2^n > n$  for all positive integers  $n$ .

**Answer:** Proof by induction

**Method:** Base case  $n = 1$ :  $2^1 = 2 > 1$ , true.

Inductive step: Assume  $2^k > k$ . Then  $2^{k+1} = 2 \cdot 2^k > 2k = k + k \geq k + 1$  (since  $k \geq 1$ ). Thus  $2^{k+1} > k + 1$ . By induction,  $2^n > n$  for all  $n \geq 1$ .

**Investigate further:** Where was the assumption  $k \geq 1$  used in the inequality chain?

6. Assuming  $k! > 2^k$  for some  $k \geq 4$ , prove  $(k+1)! > 2^{k+1}$ .

**Answer:**  $(k+1)! = (k+1)k! > (k+1)2^k > 2 \cdot 2^k = 2^{k+1}$

**Method:** Since  $k \geq 4$ ,  $k+1 \geq 5 > 2$ . Thus  $(k+1)! = (k+1)k! > 2 \cdot k! > 2 \cdot 2^k = 2^{k+1}$ .

**Investigate further:** Why was  $k \geq 4$  essential to replace  $k+1$  with  $> 2$ ?

7. Prove by induction that  $5^n - 1$  is divisible by 4 for all positive integers  $n$ .

**Answer:** Proof by induction

**Method:** Base case  $n = 1$ :  $5^1 - 1 = 4$ , divisible by 4.

Inductive step: Assume  $5^k - 1 = 4m$ . Then  $5^{k+1} - 1 = 5 \cdot 5^k - 1 = 5(4m + 1) - 1 = 20m + 4 = 4(5m + 1)$ , which is divisible by 4. By induction,  $4 \mid (5^n - 1)$  for all  $n \geq 1$ .

**Investigate further:** How can this also be proved using modular arithmetic ( $5 \equiv 1 \pmod{4}$ )?

8. Assuming  $\sum_{i=1}^k 2^i = 2^{k+1} - 2$ , show that  $\sum_{i=1}^{k+1} 2^i = 2^{k+2} - 2$ .

**Answer:**  $\sum_{i=1}^{k+1} 2^i = (2^{k+1} - 2) + 2^{k+1} = 2 \cdot 2^{k+1} - 2 = 2^{k+2} - 2$

**Method:** Add  $2^{k+1}$  to both sides:  $(2^{k+1} - 2) + 2^{k+1} = 2(2^{k+1}) - 2 = 2^{k+2} - 2$ .

**Investigate further:** What is the base case value for  $n = 1$ ?

9. Prove by induction that the sum of the interior angles of a convex  $n$ -gon is  $(n - 2) \times 180^\circ$  for  $n \geq 3$ .

**Answer:** Proof by induction

**Method:** Base case  $n = 3$ : A triangle has angle sum  $(3 - 2) \times 180^\circ = 180^\circ$ , true.

Inductive step: Assume a convex  $k$ -gon has angle sum  $(k - 2) \times 180^\circ$ . Adding a vertex to form a  $(k + 1)$ -gon adds a triangle whose interior angles sum to  $180^\circ$ . Thus the new sum is  $(k - 2) \times 180^\circ + 180^\circ = (k - 1) \times 180^\circ = ((k + 1) - 2) \times 180^\circ$ .

**Investigate further:** Why does splitting a polygon by a diagonal create a triangle and a  $(k)$ -gon?

10. Show that if  $P(k) \implies P(k + 2)$  is proved, established base cases for both  $n = 1$  and  $n = 2$  are sufficient to prove  $P(n)$  for all  $n \geq 1$ .

**Answer:** Proof by strong induction / parity split

**Method:** Base case  $n = 1 \implies P(1), P(3), P(5), \dots$  (all odd positive integers). Base case  $n = 2 \implies P(2), P(4), P(6), \dots$  (all even positive integers). Together, every positive integer is covered.

**Investigate further:** Generalise: How many base cases are needed if the inductive step is  $P(k) \implies P(k + m)$ ?

## Section C

1. A student attempts to prove  $P(n)$ : “ $n^2 + 5n + 1$  is even” for all  $n \geq 1$ . (after TMUA 2022 Paper 2 Q14)

**Answer:** B)  $P(n)$  is false for all  $n \geq 1$  despite the inductive step  $P(k) \implies P(k + 1)$  being completely valid.

**Method:** The inductive step  $P(k) \implies P(k + 1)$  is mathematically valid:  $(k + 1)^2 + 5(k + 1) + 1 = (k^2 + 5k + 1) + 2k + 6 = 2m + 2(k + 3) = 2(m + k + 3)$ , which is even if  $P(k)$  is even. However, the base case  $P(1) = 7$  is odd (false). In fact,  $n^2 + 5n + 1 = n(n + 5) + 1$  is always odd (even + odd = odd) for all  $n \geq 1$ . Thus  $P(n)$  is false for all  $n \geq 1$  despite an intact inductive step!

**Investigate further:** Why is an unanchored inductive step insufficient to prove a statement?

2. Consider the claim  $P(n)$ : “ $2^n > n^2$ .” (after SMC 2018 Q22)

**Answer:** A)  $n = 1$  and all  $n \geq 5$

**Method:** Testing small cases:  $n = 1 \implies 2 > 1$  (True);  $n = 2 \implies 4 > 4$  (False);  $n = 3 \implies 8 > 9$  (False);  $n = 4 \implies 16 > 16$  (False);  $n = 5 \implies 32 > 25$  (True). For  $n \geq 5$ ,  $2^{k+1} = 2 \cdot 2^k > 2k^2 = k^2 + k^2 > k^2 + 2k + 1 = (k+1)^2$  (since  $k^2 > 2k + 1$  for  $k \geq 5$ ). Thus  $2^n > n^2$  holds for  $n = 1$  and all  $n \geq 5$ .

**Investigate further:** Why did the inequality fail for  $n = 2, 3, 4$ ?

3. A student claims: “To prove  $P(n)$  for all  $n \geq 1$ , it is sufficient to prove  $P(1)$  and  $P(k) \implies P(2k)$  for all  $k \geq 1$ .”

**Answer:** B) False, because this only proves  $P(n)$  for  $n$  of the form  $2^m$  ( $m \geq 0$ ).

**Method:**  $P(1) \implies P(2) \implies P(4) \implies P(8) \dots$  only proves  $P(n)$  for powers of 2. Non-power-of-two integers (like  $n = 3, 5, 6, 7$ ) are never reached.

**Investigate further:** What additional step (such as  $P(k) \implies P(k-1)$ ) would complete Cauchy induction?

4. Consider the Fibonacci sequence  $F_1 = 1, F_2 = 1, F_3 = 2, F_4 = 3, \dots$  defined by  $F_{n+1} = F_n + F_{n-1}$ . What is the value of  $\sum_{i=1}^n F_i$ ?

**Answer:** B)  $F_{n+2} - 1$

**Method:** Base case  $n = 1$ :  $\sum_{i=1}^1 F_i = F_1 = 1$ ;  $F_{1+2} - 1 = F_3 - 1 = 2 - 1 = 1$ , true.

Inductive step: Assume  $\sum_{i=1}^k F_i = F_{k+2} - 1$ . Then  $\sum_{i=1}^{k+1} F_i = (F_{k+2} - 1) + F_{k+1} = (F_{k+2} + F_{k+1}) - 1 = F_{k+3} - 1 = F_{(k+1)+2} - 1$ .

**Investigate further:** Verify for  $n = 4$ :  $1 + 1 + 2 + 3 = 7 = F_6 - 1 = 8 - 1 = 7$ .

5. A student attempts to prove: “For all positive integers  $n$ ,  $n! \leq 2^n$ .”

**Answer:** B)  $n! \leq 2^n$  for  $n \in \{1, 2, 3\}$ , but  $n! > 2^n$  for all  $n \geq 4$ .

**Method:** Testing  $n = 1 : 1 \leq 2$ ;  $n = 2 : 2 \leq 4$ ;  $n = 3 : 6 \leq 8$ ;  $n = 4 : 24 > 16$ . For  $n \geq 4$ ,  $(k+1)! = (k+1)k! > 2 \cdot 2^k = 2^{k+1}$  by induction. Thus  $n! > 2^n$  for all  $n \geq 4$ .

**Investigate further:** At what rate does  $n!$  grow compared to exponential functions  $a^n$ ?

6. Consider the sum  $S_n = \frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \dots + \frac{1}{n(n+1)}$ . Using mathematical induction, what is the closed-form expression for  $S_n$ ?

**Answer:** A)  $\frac{n}{n+1}$

**Method:** Base case  $n = 1$ :  $S_1 = \frac{1}{2} = \frac{1}{1+1}$ .

Inductive step: Assume  $S_k = \frac{k}{k+1}$ . Then  $S_{k+1} = \frac{k}{k+1} + \frac{1}{(k+1)(k+2)} = \frac{k(k+2)+1}{(k+1)(k+2)} = \frac{k^2+2k+1}{(k+1)(k+2)} = \frac{(k+1)^2}{(k+1)(k+2)} = \frac{k+1}{(k+1)+1}$ .

**Investigate further:** How can this sum also be evaluated using partial fractions telescoping?

7. To prove  $P(n)$  for all integers  $n \geq 1$  using Strong Induction, what is the inductive hypothesis step?

**Answer:** B) Assume  $P(1), P(2), \dots, P(k)$  are all true, and deduce  $P(k+1)$ .

**Method:** In strong induction (complete induction), the inductive hypothesis assumes the statement holds for all values from the base case up to  $k$ , i.e.,  $P(1) \wedge P(2) \wedge \dots \wedge P(k) \implies P(k+1)$ .

**Investigate further:** When is strong induction required instead of weak induction?

8. For which positive integers  $n$  is  $n^2 < 2^n$  false?

**Answer:** A)  $n = 2, 3, 4$

**Method:**  $n = 1 \implies 1 < 2$  (True).  $n = 2 \implies 4 < 4$  (False,  $4 = 4$ ).  $n = 3 \implies 9 < 8$  (False).  $n = 4 \implies 16 < 16$  (False,  $16 = 16$ ).  $n = 5 \implies 25 < 32$  (True). Thus  $n^2 < 2^n$  is false for  $n \in \{2, 3, 4\}$ .

**Investigate further:** What is the set of integer solutions to  $n^2 = 2^n$ ?

## Section D

1. Prove by induction: Bernoulli's Inequality states that for all  $x > -1$  and all positive integers  $n$ ,  $(1+x)^n \geq 1+nx$ . At which step in the inductive transition  $(1+x)^k \geq 1+kx \implies (1+x)^{k+1} \geq 1+(k+1)x$  is the condition  $x > -1$  strictly required?

**Answer:** B) When multiplying both sides of  $(1+x)^k \geq 1+kx$  by  $(1+x)$  without flipping the inequality sign.

**Method:** Multiplying an inequality by  $(1+x)$  preserves the inequality sign if and only if  $1+x > 0 \iff x > -1$ . If  $x < -1$ , then  $1+x < 0$ , which would flip  $\geq$  to  $\leq$ , breaking the inductive transition.

**Investigate further:** What happens when  $x = -1$ ?

2. Prove by induction that  $a^n - b^n$  is divisible by  $a - b$  for all positive integers  $n$ , where  $a, b$  are distinct integers. Show the inductive step using the identity  $a^{k+1} - b^{k+1} = a(a^k - b^k) + b^k(a - b)$ .

**Answer:** Proof by induction

**Method:** Base case  $n = 1$ :  $a^1 - b^1 = a - b$ , divisible by  $a - b$ .

Inductive step: Assume  $a^k - b^k = m(a - b)$  for some integer  $m$ .

Then  $a^{k+1} - b^{k+1} = a(a^k - b^k) + b^k(a - b) = a \cdot m(a - b) + b^k(a - b) = (am + b^k)(a - b)$ . Since  $am + b^k \in \mathbb{Z}$ ,  $a - b$  divides  $a^{k+1} - b^{k+1}$ . By induction,  $(a - b) \mid (a^n - b^n)$  for all  $n \geq 1$ .

**Investigate further:** How does this generalize the algebraic identity  $a^n - b^n = (a - b)(a^{n-1} + a^{n-2}b + \dots + b^{n-1})$ ?

3. Consider a grid of  $2^n \times 2^n$  squares with one single  $1 \times 1$  square removed. Prove by mathematical induction that for every integer  $n \geq 1$ , the remaining region can be completely tiled by L-trominoes (shapes made of 3 squares forming an L).

**Answer:** Proof by induction

**Method:** Base case  $n = 1$ : A  $2 \times 2$  grid with 1 square removed leaves 3 squares in an L-shape, which is 1 L-tromino. True.

Inductive step: Divide a  $2^{k+1} \times 2^{k+1}$  grid into four  $2^k \times 2^k$  quadrants. One quadrant contains the missing square. Place 1 L-tromino at the center intersection so that it covers 1 square from each of the other three quadrants. Now each of the four  $2^k \times 2^k$  quadrants has exactly 1 square removed. By IH, each quadrant can be tiled by L-trominoes. Thus the full grid is tiled.

**Investigate further:** Why must the grid dimensions be powers of 2 for this center-tiling strategy to work?

4. Let  $a_1 = 1$  and  $a_{n+1} = \sqrt{2 + a_n}$  for  $n \geq 1$ . (a) Prove by induction that  $a_n < 2$  for all  $n \geq 1$ . (b) Prove by induction that  $a_n < a_{n+1}$  for all  $n \geq 1$ .

**Answer:** (a)  $a_n < 2$  for all  $n \geq 1$ . (b)  $a_n < a_{n+1}$  for all  $n \geq 1$ .

**Method:** (a) Base case  $n = 1$ :  $a_1 = 1 < 2$ . IH:  $a_k < 2 \implies a_{k+1} = \sqrt{2 + a_k} < \sqrt{2 + 2} = 2$ .

(b) Base case  $n = 1$ :  $a_1 = 1, a_2 = \sqrt{3} > 1$ , true. IH:  $a_{k-1} < a_k \implies 2 + a_{k-1} < 2 + a_k \implies \sqrt{2 + a_{k-1}} < \sqrt{2 + a_k} \implies a_k < a_{k+1}$ .

Since  $a_n$  is bounded above and monotonically increasing,  $\lim_{n \rightarrow \infty} a_n = L$  exists, where  $L = \sqrt{2 + L} \implies L^2 - L - 2 = 0 \implies L = 2$ .

**Investigate further:** What is the value of  $\sqrt{2 + \sqrt{2 + \sqrt{2 + \dots}}}$ ?

5. Prove by induction that  $\sum_{i=1}^n i^3 = (\sum_{i=1}^n i)^2 = \left[ \frac{n(n+1)}{2} \right]^2$  for all positive integers  $n$ .

**Answer:** Proof by induction

**Method:** Base case  $n = 1$ :  $1^3 = 1^2 = 1$ , true.

Inductive step: Assume  $\sum_{i=1}^k i^3 = \left[ \frac{k(k+1)}{2} \right]^2$ .

Then  $\sum_{i=1}^{k+1} i^3 = \left[ \frac{k(k+1)}{2} \right]^2 + (k+1)^3 = (k+1)^2 \left[ \frac{k^2}{4} + (k+1) \right] = (k+1)^2 \left[ \frac{k^2 + 4k + 4}{4} \right] = (k+1)^2 \left[ \frac{(k+2)^2}{4} \right] = \left[ \frac{(k+1)(k+2)}{2} \right]^2$ . By induction, the identity holds for all  $n \geq 1$ .

**Investigate further:** This is Nicomachus's theorem. How can it be visualized geometrically using nested square blocks?